

XANDER DUKAS

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Seeking a Mechanical Engineering Co-op — Available Jul 2027 – Dec 2027

EDUCATION

Northeastern University

B.S. in Mechanical Engineering — Minors in Aerospace Engineering & Computer Science

GPA: 3.848 / 4.0 · Dean's List (every semester) · SAT: 1510 (superscore)

Relevant Coursework: Thermodynamics, Fluid Mechanics, Statics & Mechanics of Materials, Materials Science, Differential Equations, Object-Oriented Programming

Boston, MA

Expected May 2029

Norfolk Academy

Head of the Honor Council · Head of the Robotics Team

Norfolk, VA

Graduated May 2025

EXPERIENCE

Engineering Intern — Controls Corporation of America

Jun 2026 – Aug 2026

Virginia Beach, VA — gas pressure regulators (pressure & flow systems)

- Designed a half-inch, 4,000+ psi-rated master valve to replace a purchased forged component, cutting unit cost from \$500 (purchased) to \$100 (produced in-house).
- Validated the design against CONCOA drawings and tolerances using seat-force calculations and thread specifications, matching flow requirements to established prior designs.
- Produced full SolidWorks models, engineering drawings, and manufacturing documentation (including part numbers) to support production release.
- Worked directly with CNC machinists and the manufacturing engineering team to incorporate tooling and fabrication constraints and efficiencies into the design.

PROJECTS

Mars Rover Competition — Mechanical Lead, Drone Subteam

Sep 2025 – Present

- Lead mechanical design for the drone subteam; developing a new bicopter configuration with maximized propeller size to increase efficiency under the competition weight limit.
- Previously designed the 1.5 kg dummy-weight payload for the team's 5 kg competition drone, optimizing center-of-mass placement and a connect/disconnect mechanism that reduced payload oscillation during flight and release.
- Built a complete drone in an introductory drone program (Sep 2025 – Dec 2025) with guidance from the subteam head; learned ArduPilot flight-controller setup and soldering.
- Team placed 6th in the Mars Rover Competition.

RxManager — IoT Medication Dispenser github.com/Geordorini/rxmanager

Jan 2026 – May 2026

- Built an IoT pill dispenser (ESP32, Supabase, Vercel dashboard) that releases a pill only when the user presses a button, logging which medication, when, and how many; the dashboard flags missed doses for patients, caregivers, and doctors.
- Directed AI coding tools (Claude, ChatGPT) to build the dashboard front end, Supabase-backed API/back end, and ESP32/MicroPython firmware.

Ride Engineering Competition — Theme Park Engineering Club

Jan 2026 – Present

- Designed and analyzed an amusement ride to ASTM F2291, the industry standard practice for the design of amusement rides and devices; placed 2nd overall in the competition.

Boat Design Challenge — High School Engineering Class

Mar 2025 – Apr 2025

- Built a boat hull from wood and tarp with a 3D-printed propeller; won the class tournament.

TECHNICAL SKILLS

CAD: SolidWorks, AutoCAD, OnShape

Programming & Analysis: MATLAB, Python, MicroPython

Certifications: SolidWorks CSWA (Certified Associate)

Fabrication: 3D printing, soldering, manual milling

Tools: Git, Supabase

LEADERSHIP & ACTIVITIES

Cheese Club

Sep 2025 – Present

- Event Coordinator, organizing club collaboration events across campus; previously handled cheese acquisition for the club.
- Theme Park Engineering Club** — Member
- Mars Rover Competition Team** — Mechanical Lead, Drone Subteam